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Contributions of Universities to Regional Economic Development: A Quasi-experimental Approach

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GOLDSTEIN H. A. and RENAULT C. S. (2004) Contributions of universities to regional economic development: a quasi-experimental approach, *Regional Studies* **38**, 733–746. Universities potentially contribute to regional economic development in a number of ways: research, creation of human capital through teaching, technology development and transfer, and co-production of a favourable milieu. We find that the research and technology creation functions generate significant knowledge spillovers that result in enhanced regional economic development that otherwise would not occur. Yet, the magnitude of the contribution that universities' research and technology development activities play is small compared with other factors. We use a quasi-experimental approach to explain variation in change in average earnings per job across 312 metropolitan statistical areas in the USA in 1969–86 and 1986–98.

Regional economic development   Higher education   Universities   Knowledge spillovers

GOLDSTEIN H. A. et RENAULT C. S. (2004) La contribution des universités au développement économique régional: une façon quasi-expérimentale, *Regional Studies* **38**, 733–746. Potentiellement, les universités contribuent de plusieurs façons au développement économique régional: à savoir, la recherche, la création du capital humain à travers l'enseignement, le développement et le transfert de la technologie, et la coproduction d'un milieu propice. Il s'avère que les fonctions recherche et technologie engendrent d'importantes retombées de connaissances qui permettent un développement économique régional accru qui n'aurait pas vu le jour dans d'autres circonstances. Toujours est-il que la contribution des actions des universités dans les domaines de la recherche et de la technologie s'avère peu importante par rapport à celle des autres facteurs. Une façon quasi-expérimentale sert à expliquer la variation des gains moyens par emploi à travers 312 MSA aux Etats-Unis de 1969 à 1986 et de 1986 à 1998.

Développement économique régional   Enseignement supérieur   Universités   Retombées de connaissances

GOLDSTEIN H. A. und RENAULT C. S. (2004) Beiträge der Universitäten zur regionalwirtschaftlichen Entwicklung: ein quasi experimentaler Ansatz, *Regional Studies* **38**, 733–746. Universitäten tragen potentiell auf mancherlei Art und Weise zur regionalwirtschaftlichen Entwicklung bei: durch Forschung, Schaffung von Menschenkapital mittels Unterricht, Entwicklung und Übertragung von Technologie, sowie Miterzeugung eines günstigen Milieus. Es wird festgestellt, daß Schaffensfunktionen von Forschung und Technologie signifikante Wissensüberschüsse erzeugen, die sich in qualitativ verbesserter regionalwirtschaftlicher Entwicklung niederschlagen, die sonst nicht stattfinden würde. Doch der Umfang der von Universitäten geleisteten Tätigkeit auf den Gebieten der Forschungs-und Technologieentwicklung ist im Vergleich mit anderen Faktoren nur gering. Es wird ein quasi experimenteller Ansatz dazu benutzt, Abweichungen bei Veränderungen durchschnittlicher Einkommen pro Erwerbstätigkeit in 312 MSAs der Vereinigten Staaten in den Perioden 1969–86 und 1986–98 zu erläutern.

Regionalwirtschaftliche Entwicklung   Höhere Bildung   Universitäten   Wissensüberschüsse

GOLDSTEIN H. A. y RENAULT C. S. (2004) Las contribuciones de las universidades al desarrollo económico regional: un enfoque cuasi-experimental, *Regional Studies* **38**, 733–746. Las universidades contribuyen potencialmente al desarrollo económico regional de varias formas: investigación, creación de capital humano por medio de la enseñanza, desarrollo y transferencia de tecnología, y co-producción de un entorno favorable. Encontramos que las funciones de investigación y de creación de tecnología generan desbordamientos de conocimiento significativos que resultan en una mejora del desarrollo económico regional, que de lo contrario no ocurriría. Aún así, la magnitud de la contribución que desempeñan las actividades de

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investigación y de desarrollo de tecnología es pequeña en comparación a otros factores. Nosotros utilizamos un enfoque cuasi-experimental para explicar las variaciones que se producen en el promedio de ingresos por empleo a través de 312 MSAs en los Estados Unidos en los periodos comprendidos entre 1969–86 y 1986–98.

Desarrollo económico regional Enseñanza superior Universidades Desbordamientos de conocimiento

JEL classifications: O18, R11

INTRODUCTION

Since the mid-1980s, academic interest in the relationship between knowledge production within a region and the region's economic growth and development performance and prospects has burgeoned. The reasons for this increased interest include dramatic changes in the global economy and conditions of regional competitiveness, the increased importance of knowledge inputs in the production of a wide range of goods and services, and more recently with advances in 'the new growth theory'.

The relatively severe economic downturn of 1981–82 hit the traditional manufacturing sectors particularly hard. In response, economic development officials at the state and regional levels began investing in a variety of programmes and institutions aimed at strengthening the knowledge infrastructure of their regions. Public universities were often the primary institutional recipient for many of the investments in the knowledge infrastructure. Faced with budgetary squeezes, leaders at many public universities were only too eager to accept responsibility (and the funding that came with it) for adding economic development to their traditional missions of teaching, research and public service. The passage of the Bayh–Dole Act of 1980 gave universities an even larger incentive to engage in entrepreneurial activities by giving them intellectual property rights on patentable inventions originally stemming from federal government-sponsored research.

The purposes of these public investments in the knowledge infrastructure have varied among states and regions depending upon particular economic conditions and perceived needs. Generally, however, they have focused on attracting, nurturing and retaining high-tech industries and innovative firms to provide existing residents with the range of skills and competencies they will need to be productive in the knowledge economy, and to help the region sustain its competitiveness in the future. Beliefs about the importance of such investments tended to be based upon neither sound empirical evaluations nor theoretical arguments. Instead, they emanated from a few well-known and celebrated 'success' cases such as Silicon Valley in California, Route 128 in Massachusetts and the Research Triangle in North Carolina, USA: 'We can be the region spawning the next Apple Computer'. There was also a tendency for state and regional development officials to play 'follow the leader', so as not to risk being the one left behind (ATKINSON, 1989).

For any significant public investment, there are always questions about the magnitude of impacts, their distribution and the investment's effectiveness in achieving the desired objectives. In the particular case of investments in the production of knowledge and the knowledge infrastructure, the expected regional economic impacts revolve around the nature of spatial spillovers and other forms of externalities generated from knowledge-producing organizations. If one focuses on universities as the dominant type of knowledge-producing organizations,[1] the following questions should be addressed:

- To what extent do institutions of higher education, and research universities specifically, generate regional economic development outcomes that would otherwise not occur?
- Which university-based activities, e.g. teaching, basic research, extension and public service, technology transfer, technology development, businesses spinning off from university research, etc., are most responsible for any net regional economic development impacts from the presence of universities?
- Through what mechanisms or channels does knowledge production (broadly considered) within universities lead to economic development outcomes in the surrounding region? Is it, for example, through economic transactions between actors or units within the university and external organizations, through spillovers or through milieu effects, which are particular kinds of localization economies?
- What are the critical internal and external factors that condition the contribution of knowledge-producing organizations to regional economic development and determine the share of total economic development impacts retained within the region vis-à-vis the rest of the world?

The present paper focuses on addressing the first, second and fourth questions. The empirical analysis employs a quasi-experimental design with a large sample of metropolitan regions in the USA. The results provide some *indirect* insights on the third question: the modes of transmission of impacts from universities to the larger region.

The paper is organized as follows. The next section lays out conceptually the different ways universities potentially can contribute to regional economic development. The third section provides a brief, critical review of the two primary methodological approaches

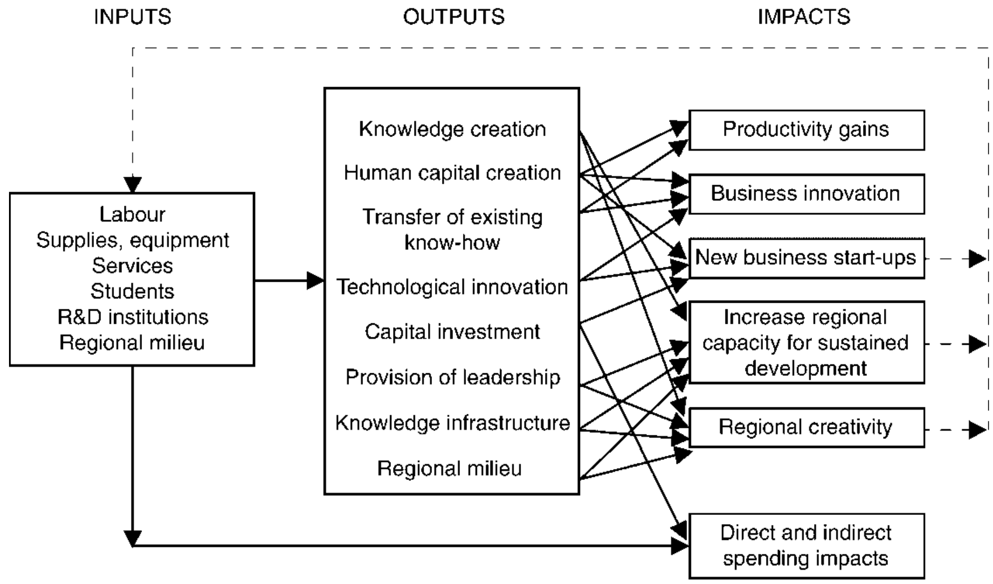

Fig. 1. *University outputs and expected economic impacts.*
Source: GOLDSTEIN et al. *(1995)*

used to estimate university impacts on regional economic development, and it introduces the advantages of using a quasi-experimental design. The fourth section describes the study population, the measures, the data, the hypotheses and the models used, while the empirical results are presented in the fifth section. The sixth section focuses on university spillovers to their regional environment, given the empirical results.

OUTPUTS AND POTENTIAL IMPACTS OF RESEARCH UNIVERSITIES

The literature on the economics of knowledge production and innovation, on knowledge spillovers, and on the changing role of institutions of higher education suggests there is a wide range of ways that universities potentially can contribute directly and indirectly to regional economic development. The modern research university, at least in the USA and increasingly in Western Europe, can be considered as a multiproduct organization. Within legal, cultural and political constraints, the research university can choose its optimal product mix in response to perceived changes in its markets and the availability and prices of its inputs. (Unlike corporate research and development (R&D) organizations, universities do *not* consider changing locations as responses to changing market conditions.)

GOLDSTEIN *et al.* (1995) synthesize a wide range of the literature on institutions of higher education, and identify and describe the range of products, or outputs, from modern research universities. They suggest how each type of outputs might *potentially* lead to specific economic development impacts. The outputs include: (1) knowledge creation, (2) human capital creation, (3) transfer of existing know-how, (4) technological innovation, (5) capital investment, (6) provision of regional leadership, (7) co-production of the knowledge infrastructure and (8) co-production of a particular type of regional milieu. The potential impacts include: productivity gains, business innovation, new business start-ups, an increase in regional economic development *capacity* (for sustained, long-term development), regional creativity, and direct and indirect spending impacts. For the hypothesized relationships between the university outputs and the hypothesized regional development impacts, see Fig. 1.

AVAILABLE METHODOLOGICAL APPROACHES

The literature suggests there are two primary ways to investigate the link between knowledge production and regional economic development: the case study, and econometric models of knowledge production and spillovers.

Case studies

One can trace case studies of the economic impact of specific universities in the published literature at least to the early 1970s. These case studies attempt to estimate the contribution of a particular institution of higher education to the regional economy in one of two typical ways. In the first, a regional input–output model is used to estimate the direct and indirect spending effects of the organization on the region's output, earnings and employment. In the second, a sample of other businesses or organizations in the region are asked in a questionnaire what the perceived importance of the university was to the location decision, productivity and competitiveness, innovation, and output levels of the respondent's organization.

THANKI (1999) reviews much of the case study literature, particularly in the experience of the UK, and finds that such studies to date have *not* captured many of the potential ways that institutions of higher education can lead to regional economic development, either because of a too narrow understanding of how regional development occurs, or because of the limitations of the techniques used. For instance, regional input–output models and other multiplier techniques can only capture economic growth that stems from backward linkages induced by the spending of an institution of higher education. The reliance only on such techniques fails to take into account the unique qualities of knowledge as a product of such organizations, essentially treating a university no differently from any other kind of organization that hires and pays labour and purchases supplies and equipment from both regional and outside sources. FELSENSTEIN (1994a) reviews a large number of economic impact studies conducted in the USA using input–output and related multiplier techniques. For the most part, these studies suffer from the same set of limitations as discussed by THANKI (1999).

Within the case study framework, a more eclectic set of data collection and analytic techniques has been used to try to estimate a wider range of putative regional economic effects of universities, in addition to those measured by input–output models. Several researchers (SMILOR *et al.*, 1990; BRETT *et al.*, 1991; KO, 1993; STEFFENSEN *et al.*, 2000) focus on 'counting' the number of spin-offs from university research centres and academic departments, using newspapers and the regional business press, as well as personal interviews as sources. LUGER and GOLDSTEIN (1991) and GOLDSTEIN and LUGER (1992, 1993) use targeted questions sent in a questionnaire to a stratified sample of high-tech companies recently located in a region to estimate to what extent the region's research universities were an important factor in their regional location decision. By asking the counterfactual 'If the University of ... was not in the region, how likely would you have located your business in this region: (1) Very likely, (2) Likely, (3) Perhaps, (4) Unlikely, (5) Very unlikely', and then applying a specific probability to each response, the researchers estimate the total employment and output in the region's high-tech sector that otherwise would not have been located there.

GOLDSTEIN and LUGER (1992) also estimate the human capital impacts on a region of a university by linking historical student registration and alumni records to estimate the per cent of graduates, by degree and discipline, who have remained and been employed within the region over time. FELSENSTEIN (1994b) estimates the amount of university-induced migration to a region (that otherwise would not have occurred) using Keynesian-type multipliers.

SAXENIAN's (1994) comparative case studies of Silicon Valley and Route 128, while not attempting to measure the impact of specific universities on regional development, tries to explain the difference in economic performance and success between the two areas by using ethnographic techniques focusing on the extent and qualities of inter-organizational linkages and collaborations within the respective regions.

In summary, the principal strengths of the case study approach are: (1) its flexibility in collecting *primary data* through surveys and interviews, and documentary evidence in the form of internal reports, in order to measure, or count, particular kinds of university outputs and some of the associated impacts; (2) the generation of information about the internal organization and culture of a particular university in order to relate these internal attributes to effectiveness in contributing to regional economic development; and (3) generating information about the soft and 'fuzzy' side of how universities can stimulate economic development through the provision of leadership and the co-production of a creative milieu, concepts that are difficult to measure. The often intensive data-gathering effort to estimate the spending impacts of the university on the regional economy with the use of input–output and related multiplier techniques is almost always conducted within a case study approach. The case study approach is most appropriate for counting or estimating incidences of technology transfer, technological innovation (e.g. in the form of patents issued and licenses sold), capital investment and the number of business spin-offs.

The principal weakness of the case study approach is solving the 'attribution' problem, i.e. controlling for other putative factors when trying to make causal inferences between university activities and regional economic development outcomes. This is most problematic for estimating *indirect impacts* such as regional productivity gains, increases in innovative activity in the region, increased economic development capacity and regional creativity. In principle, one could conduct multiple cases studies using a standard set of data and techniques in order to mitigate the inability statistically to control for other intervening factors, but this is likely to be impractical owing to the relatively large cost of conducting individual case studies. In several of the case studies discussed above, researchers have asked respondents about the 'perceived' impacts or the 'perceived importance' of the university, or have constructed counterfactuals, but the use of such data and techniques carries with it some obvious and significant validity threats.

Econometric models of knowledge production and spillovers

More recently, economists have tried to assess the economic impact of new innovation such as that produced at a university by using econometric models. Early models focus on production functions that contained an R&D variable. GRILICHES (1979) proposes a specific knowledge production function assumed to be Cobb–Douglas in form, relating *Y*, some output, to labour, capital and knowledge, measured by R&D expenditures. This work was based on the empirical work of SOLOW (1957) that shows the existence of a latent variable besides capital and labour in aggregate production functions, and ARROW (1962), who describes the impact of learning (gaining knowledge) on the production function.

JAFFE (1989) modifies Griliches' knowledge production function to the form generally used by researchers today:

$$\log K = \alpha_0 + \alpha_1 \log \mathrm{RD} + \alpha_2 \log \mathrm{URD} + \varepsilon \qquad (1)$$

where *K* is some measure of innovation, RD is industry R&D expenditure and URD is university R&D expenditure. Jaffe finds geographically mediated spillovers from university research to commercial innovation. A significant effect of university research on corporate patents is found, particularly in drugs, medical technology, electronics, optics and nuclear technology. He states that university research appears to have an indirect effect on local innovation by inducing industrial R&D spending.

A number of researchers modify and expand this model. JAFFE *et al.* (1993) follow this work by looking at the geographical location of patent citations to understand the extent to which knowledge spillovers are localized geographically. They find that citations to domestic patents are more likely to be domestic, and more likely to come from the same state and metropolitan statistical area (MSA) as the cited patents. ACS *et al.* (1994) use the Griliches–Jaffe model to understand the degree to which university and corporate R&D spills over to small firms as compared with large firms. They find that university spillover was more important to small firms than to large firms. VARGA (2000) uses the Griliches–Jaffe knowledge production function to try to model agglomeration. He develops a hierarchical linear model by modelling each α to include the dependence of the knowledge transfers on the concentration of high technology production and business services.

KEILBACH (2000) takes a different path. Incorporating the work of KRUGMAN (1991) and FUJITA *et al.* (1999), he builds a dynamic model of regions using the endogenous growth model including human capital and knowledge spillovers to separate the effects of spillover and agglomeration. The model builds a class of discrete dynamic systems, an *n*-dimensional array of sites, each with local rules and global rules. Assuming the initial endowments of capital and labour are identical in each region and that the distribution of knowledge is random, that capital moves globally, that labour migrates locally but that human capital spills over, Keilbach observes agglomeration under assumptions of increasing returns only to R&D. The model demonstrates a self-reinforcing structure where labour agglomerates no matter what, but agglomerates quicker with increasing returns to R&D.

The various authors above, as well as others, have progressed considerably the understanding of the nature of knowledge spillovers. These studies demonstrate that knowledge, as an output of an organization, has unique qualities that render knowledge-producing organizations *not* just like any other large organization that purchases inputs locally and employs a large number of people. Rather, there are significant external benefits of knowledge production in the form of spatial spillovers that lead to higher levels of innovative activity among other firms within the region.

There are, however, some limitations in the use of this approach for estimating the contribution of universities to regional economic development. These include measurement and data issues, the limited scope of potential outputs and impacts stemming from universities, and thus the limited ability to separate different ways that universities contribute to regional economic development.

The primary challenge with using these types of models is how to measure the dependent variable: innovation. ACS *et al.* (1994, p. 337) quote Griliches as saying, 'The dream of getting hold of an output indicator of inventive activity is one of the strong motivating forces for economic research in this area'. Since it is unknown how to measure innovation directly, proxy indicators, each with their own inherent challenges, are used.

VARGA (2000) and ACS *et al.* (1994) use a one-time cross-section data set from the Small Business Administration on new product introductions. They reason that these innovations had market value since companies introduced them and that this was a better measure than patents or other pre-market indicators. From a company point of view, product introductions are still indicators of 'probable' market value. Until sales of a product are known, the market value is still a prediction. So, while it is true that product introductions, being further down the product development chain than patents, are probably better indicators of innovations with market value, they are still predictors, not measures. The other significant issue with this data set is that it is a one-time indicator constructed in 1982 and now highly dated.

JAFFE (1989) and JAFFE *et al.* (1993) use the number of patents as their dependent variable. There are many issues with this indicator, as Jaffe recognizes. For instance, there are factors outside knowledge generation that affect the number of patents that might arise from a given piece of R&D.

First, not all innovation is patentable. Other types of innovation not included in patents are codified knowledge embodied in copyrights, trade secrets and scientific papers, as well as tacit know-how and shared expertise. Since the transmission of tacit information is an important explanation of why spillover has local effects (as compared with patents that can be licensed worldwide), the inability to measure this dimension is a significant handicap.

Second, especially within universities, changes in patenting policies and the university's fiscal environment significantly affect the number of patents applied for and granted. THURSBY and THURSBY (2000), for instance, show that the recent growth in licensing activity at US research universities was based on input growth (i.e. more disclosures) of *lowering quality* and on an increase in the propensity of administrators to patent and license faculty inventions as a revenue-raising strategy.

In fact, the general hypothesis has been that patents are a proxy for inventive output and that patent citations are a proxy for knowledge flows or spillover, the real source of innovation. Recent work (HALL *et al.*, 2000) strongly suggests that patent citations are well correlated with the market value of the knowledge and are the best indicator of innovation and knowledge spillover. To date, patent citations have not been used in the knowledge production function, although JAFFE and TRAJTENBERG (1996) use them to find that diffusion of knowledge is localized geographically.[2]

The second major limitation of this approach for the present purposes is that while it yields estimates on some of the outputs of universities, i.e. technological innovation, it does not take into account other outputs of universities discussed in the second section – human capital creation, building regional capacity, stimulating business start-ups and the regional milieu effect – that potentially have impacts on regional economic development.

QUASI-EXPERIMENTAL APPROACH

Quasi-experimental designs are best described in comparison with true experimental designs. The hallmarks of true experimental designs are that the researcher can manipulate how and which cases receive 'treatment', such that the treatment can be randomly assigned to cases drawn from a sample of a given study population. Randomization means that probabilistically there should be no systematic difference on any rival causal factors between those cases receiving the treatment and those that do not. Thus, any differences in the dependent variable between the treatment and non-treatment groups can normally be attributed to the treatment effect.

In field settings as opposed to laboratory settings, the ability of researchers to select cases randomly for treatment is rare. Instead, they must accept the naturally occurring variation in a given treatment variable, in this case the regional distribution of universities by size and type. In other cases, it might be technically feasible for researchers to manipulate the selection of cases for treatment or non-treatment, but it is ethically prohibited. Quasi-experimental designs are a class of designs in which the researcher strategically manipulates the study population, the period and the sampling to construct control groups that enable one to control for some of the rival factors that cannot be explicitly included as variables in a statistical model. COOK and CAMPBELL (1979) and SHADISH *et al.* (2002) provide comprehensive and detailed descriptions of quasi-experimental designs.

The control group cases cannot be assumed to be equivalent to the treatment cases in all respects except for the treatment variable itself. However, such selection differences can be taken into account by measuring and comparing gain scores – *differences* in the outcome variable(s) between a pre- and post-test – instead of just comparing post-test scores.

In the context of conducting research on the relationship between knowledge production in a region and regional economic development outcomes, the presence of research universities or other significant knowledge-producing organization in a region would represent the treatment variable. The dependent variable would be the gain, or difference, of some meaningful economic development outcome over a pertinent period. The control group would consist of regions without research universities (or other significant knowledge-producing organizations). Additional control groups can be selected effectively to control for other factors that might also explain variations in the dependent variable in addition to the presence or absence of a research university. Thus, one might select two treatment groups, one with small-to-medium-sized regions with research universities, the other with large-sized regions with research universities, and two control groups (small–medium without research universities, and large without research universities). One would then compare the gains for all four groups to take into account the effect of agglomeration economies and its interaction with the presence of universities.

It is important to distinguish quasi-experimental designs from more commonly used large sample cross-sectional statistical analyses. In the latter, the researcher would randomly select a large sample of regions from the full population available, enter measures on a number of variables that the literature has suggested might explain the dependent variable into a multiple regression model, and then estimate the model. The interpretation of the effect of research universities would be based upon the significance, sign and magnitude of the coefficient estimate on the measure(s) selected for the presence or level of activity of a research university in the region. In effect, the researcher statistically controls for all other factors by entering these explicitly into the model. In contrast, quasi-experimental designs first attempt to control for other putative factors by the strategic design of treatment and control groups such that for some putative causal variables there is no variation allowed. For others, there is an attempt to maximize variation. Only after these possibilities are practically exhausted does the researcher introduce statistical controls.

The standard types of statistical procedures used to make inferences about the effect of the treatment using non-equivalent control groups include analysis of covariance with multiple covariates (essentially multiple regression) and analysis of variance using gain scores, equivalent to a difference of means test with two groups. Reichart (in COOK and CAMPBELL, 1979) discusses in detail the choice of a particular statistical model for a given set of data and research situation to minimize the possibility of yielding biased results. In impact analysis studies, ISSERMAN (1987) uses the difference of means (*t*-test) to measure 'equivalence' between the experimental case and the control group before events triggering impacts and to infer the significance of the net impact of the treatment. REED and ROGERS (2003), on the other hand, assume a number of covariates and employ a multiple regression model of the form:

$$y_i = a + bD_i + f(X_i) + e_i,\ i = 1, 2, \ldots, N \qquad (2)$$

where D is a dummy variable indicating whether a given observation was 'treated' with the policy, X is a set of K variables known to be independently related to y, e is an error term, and for generality f is a non-linear function.

Although quasi-experimental designs have been used occasionally in published regional research (e.g. ISSERMAN, 1987; ISSERMAN and MERRIFIELD, 1987), to the present authors' knowledge, they have not been used in attempts to estimate the contribution of knowledge-producing organizations to regional economic development.

Study objectives, unit of analysis and study population

The objectives of the present study are (1) to estimate the magnitude of the contribution of universities to changes in regional economic well being, controlling for other factors, and (2) to try to separate the regional economic development impacts of different functions of universities.

The unit of analysis in the study is the metropolitan statistical area (MSA). All variables are measured for the MSA. The study population consists of all 312 MSAs in the USA based on their geographical definitions in the 1990 Census.[3] The temporal period is from 1969 to 1998.

Measures of regional economic development

The measure of regional economic development is average annual earnings per worker. This contrasts with many studies that use per capita income as a measure of regional economic well being. The most important difference between the two measures is that average earnings per worker takes into account only earned income, while per capita income includes unearned income including dividends, rent, interest and transfer payments. These unearned sources of income are particularly significant in regions with a large number of retirees. In addition, average earnings per worker takes into account only the economically active persons in the denominator. Finally, average earnings per worker focuses on the *quality* of jobs in a region as the most important dimension of improvement in regional economic well being.

To separate out changes in national macroeconomic conditions over the period, including changes in the value of the dollar (US$), a normalized index for each MSA is constructed by dividing the MSA's average earnings per worker by the US average earnings per worker for the same year, then multiplying the ratio by 100. Thus, an index of 110.0 for a particular MSA means that its average earnings per worker for that year is 10.0% higher than for the nation.

The dependent variable is calculated as the difference in the index for a given MSA between 2 years. Thus, if a particular MSA had an index value of 110.0 in 1969 and one of 120.0 in 1999, then the dependent variable would be 10.0. Of course, it is possible for MSAs to have negative values on the dependent variable.

One benefit of using change in the index value is that it helps control for some local factors that are endemic to particular areas. For example, in many small MSAs with large universities, the average earnings per worker is distorted downward by a relatively large number of students employed in low-wage, part-time jobs. This distortion is minimized when the difference, rather than the level, of average earnings per worker is used.

Measures of university presence

The presence of universities in an MSA with four different variables was measured to test specific hypotheses described below. The first measure is whether there is a top 50 research university within the MSA at the beginning of the respective period. A top 50 designation is based upon rankings of universities in terms of total research expenditures compiled by the National Science Foundation on an annual basis.[4] The second measure is an alternative measure of university-based R&D activity: the sum of the total research expenditures across all universities within the MSA for a given annual period.

The teaching and human capital creation activity of higher education within the MSA is measured as the sum of all degrees awarded in all institutions of higher education within the MSA for a given year. This third measure gives one the ability to separate the effect of the teaching function, and indirectly the milieu effect of universities, from its research function.

The fourth measure is the number of patents assigned to universities within the MSA. It allows one to separate the technology development function of universities from their more traditional research function.

Control variables

There are, of course, a number of other causal factors that might help to explain variation in the magnitude of change in a region's economic well being. First are *agglomeration economies*, measured by the total MSA employment at the beginning of the period. Three MSA size categories are also used: small, medium and large employment levels, partially to define control groups.[5]

Other factors include the following:

- Region of the USA (Northeast, Midwest, South, West) is used to control for broad shifts in the regional distribution of population, employment and capital investment from the Northeast and Midwest, to the South and West over much of the period. Region of the USA is measured as a dichotomous variable and entered into the statistical model as three dummy variables.
- Industry structure. Three dimensions of the industry structure of MSAs are included: per cent earnings in manufacturing, per cent earnings in business services and change in per cent earnings from manufacturing over the period. These variables take into account that concentrations of manufacturing and business may increase a region's capacity for regional economic development.
- Location and accessibility. Accessibility to markets and 'centrality' is a traditional factor used to explain regional growth. It is measured in terms of accessibility by air transportation, and specifically whether the MSA has an airport classified as a large, medium, small or no hub. If the MSA has more than one airport, it is assigned to the category of the largest airport.
- Entrepreneurial activity. Two different measures of MSA entrepreneurial activity were used. The first is the per cent of total MSA earnings generated by proprietorships, i.e. self-employed workers. This is one dimension of entrepreneurship that takes into account small business vitality. The second measure is the number of patents assigned to private-sector companies in an annual period. This takes into account the technology creation dimension of entrepreneurialism.
- Base-year level of average earnings per worker. To control for a possible endowment effect (the rich get richer), average earnings per worker are included at the start of the period.

Hypotheses and designs to test them

The first hypothesis is that research universities contribute significantly to regional economic development, controlling for other factors. This is tested in two ways. First, if the hypothesis is true, then the gain in average earnings per worker should be higher for regions with top 50 research universities than those without such institutions. Thus, the mean gain in average earnings per worker in regions with research universities is compared with the same dependent variable for regions without top 50 research universities, controlling for the size of the MSA, the region of the USA and for industry structure, and a difference of means test for statistical significance is applied. Second, in a multiple regression model, the coefficient on total university research expenditures in the region should be positive and significant, with other putative causal factors included as control variables.

The second hypothesis is that universities' technology development activity contributes significantly to regional economic development. Without a good direct measure of university technology development for all institutions of higher education within the MSA, one of the strategies of quasi-experimental designs is used, i.e. manipulation of the period. The fulltime period is split into two parts: 1969–86 and 1986–99. The rationale is that around the mid-1980s, many universities started to incorporate economic development missions, both in response to reductions in federal research support and to the Bayh–Dole Act (Feller, 1990), as well as the initiatives of many state legislatures promoting universities as economic development actors in response to the recession of 1981–82 and lagging competitiveness. Around the mid-1980s, universities began creating technology transfer offices to support patenting and licensing activities, building incubators and research parks, and, occasionally, creating or investing in venture capital funds. Thus, if universities' involvement in technology development in various forms has been an important contributor to regional economic development, then the difference in the dependent variable (gain in average earnings per worker) between regions with research universities and those without should be larger in 1986–99 compared with 1969–86.

The third hypothesis is that human capital creation (teaching) and the milieu functions are important contributors to regional economic development, controlling for the magnitude of research and technology development activity. Human capital creation and milieu together are measured by the number of degrees awarded annually by all institutions of higher education in the region. If this is true, then the coefficient in the multiple regression model will be positive and significant in both periods.

The fourth hypothesis to be investigated is whether agglomeration is more important than the presence of research universities as a contributor to regional economic development. If this is true, then the coefficient for size of MSA will be positive and significant, and stronger than coefficients on measures of university presence variables in the regression model, and more so in 1986–99 than in the earlier period.

The fifth hypothesis is a variant on the fourth: that there is *interaction* between the presence of research universities and agglomeration. More specifically, the hunch is that they are substitutes. If this is true, then the effect of research universities will be smaller the larger the MSA. This is tested by comparing the difference of means between MSAs with top 50 research universities against those without top 50 research universities by size class. Differences should be largest for the smallest class of MSAs and smallest for the largest class of MSAs. These differences should also be more prominent in 1986–99 than in 1969–86. This hypothesis is also tested by explicitly introducing an interaction variable in the regression models of total universities' R&D expenditures multiplied by MSA employment.

EMPIRICAL RESULTS

As stated above, two different types of analysis of the data were conducted. The first was a set of difference-of-means tests between MSAs with and without top 50 research universities overall, and then stratified by a number of control variables. These difference-of-means tests were done for each separate period to compare for temporal differences. The results are shown in Tables 1 and 2, respectively.[6]

There are no significant differences in the change in average earnings per job between MSAs with top research universities and those without for 1969–86. These results hold overall, and for each MSA size category, location and type of industry structure. Two-way cross-tabs combining MSA size categories, location and industry structure also showed that the presence of a research university was not associated with a change in average earnings per job, except for the one case: small MSAs with high manufacturing. Here, however, the small number of such MSAs without research universities outperformed those with top research universities.

The results for 1986–98 tell quite a different story. In the entrepreneurial university period, the differences between MSAs with and without top research universities are significant and positive for the overall study population. When one disaggregates by type of MSA, the same results are obtained for small and medium MSAs, those in the Northeast, South and West, and with both high and low percentages of manufacturing. Within small MSAs, the positive presence of research universities is most prominent for locations with low

Table 1. Change in average earnings/job, 1969–86

	Metropolitan statistical areas with RU		Metropolitan statistical areas without RU		
Type of area	Mean	*N*	Mean	*N*	Difference
All	−1.20	37	−0.54	275	−0.66
Small	−2.34	10	−1.00	140	−1.34
Medium	−1.76	19	−0.25	128	−1.51
Large	1.50	8	3.25	7	−1.75
Northeast	−0.72	6	−2.53	40	1.81
South	4.36	9	4.41	119	0.05
Midwest	−6.04	11	−5.98	66	0.06
West	−1.20	11	3.56	50	2.36
Low manufacturing	0.08	20	1.15	144	−1.07
High manufacturing	−2.73	17	−2.42	131	−0.31
Small, low manufacturing	−0.59	7	−0.55	75	−0.04
Medium, low manufacturing	−0.56	10	2.88	66	−3.44
Large, low manufacturing	3.77	3	1.16	3	2.61
Small, high manufacturing	−6.40	3	−1.51	65	−4.89**
Medium, high manufacturing	−3.09	9	−3.59	62	0.50
Large, high manufacturing	0.14	5	1.10	4	−0.96
Small, Northeast	−7.90	1	−5.90	12	2.66
Small, South	3.87	3	4.09	64	−0.07
Small, Midwest	−4.77	3	−5.52	33	0.54
Small, West	−4.27	3	−4.79	31	0.55
Medium, Northeast	2.00	2	−1.15	25	3.15**
Medium, South	5.34	5	4.48	52	0.86
Medium, Midwest	−6.51	8	−6.47	32	−0.04
Medium, West	−3.00	4	−1.57	19	−1.43
Large, Northeast	−0.13	3	−0.63	3	0.50
Large, South	0.90	1	10.17	3	−9.27
Large, Midwest		0	−5.80	1	
Large, West	2.87	4		0	

Notes: $^{*}p<0.05$, $^{**}p<0.001$.
RU, research university(ies).

Table 2. Change in average earnings/job, 1986–98

Type of area	Metropolitan statistical areas with RU Mean	N	Metropolitan statistical areas without RU Mean	N	Difference
All	3.37	36	−4.23	276	7.60**
Small	−0.89	12	−5.32	215	7.09**
Medium	5.87	12	−1.22	52	7.09*
Large	5.13	12	4.59	9	0.54
Northeast	7.36	8	−1.08	39	8.44*
South	1.89	10	−4.06	118	5.95**
Midwest	−3.08	9	−5.45	68	2.37
West	7.91	9	−5.58	52	13.49**
Low manufacturing	2.17	25	−3.71	173	5.88**
High manufacturing	6.09	11	−5.09	103	11.18**
Small, low manufacturing	−1.20	8	−4.69	137	3.49*
Medium, low manufacturing	0.26	7	−1.53	30	1.79
Large, low manufacturing	6.21	10	7.68	6	−1.47
Small, high manufacturing	−0.27	4	−6.44	78	6.17
Medium, high manufacturing	13.72	5	−0.79	22	14.51*
Large, high manufacturing	−0.25	2	−1.60	3	1.35
Small, Northeast	−3.40	1	−3.36	27	−0.04
Small, South	−2.40	3	−5.23	92	2.83
Small, Midwest	−4.42	6	−5.77	55	3.16
Small, West	13.20	2	−6.46	42	1.35
Medium, Northeast	10.62	4	3.09	10	7.53
Medium, South	2.29	4	−0.75	24	3.04
Medium, Midwest	−0.70	1	−4.16	11	3.46
Medium, West	6.47	3	−4.33	7	10.80
Large, Northeast	6.60	3	8.80	2	−2.20
Large, South	5.63	3	9.95	2	−4.32
Large, Midwest	−0.25	2	−3.80	2	3.55
Large, West	6.35	4	3.80	3	2.55

Notes: $*p<0.05$, $**p<0.001$.
RU, research university(ies).

manufacturing. In medium MSAs, the result is strongest for locations with high manufacturing.

The second type of analysis is based upon ordinary least-squares regression. Separate models are estimated for each of the two time segments. The results are shown in Tables 3 and 4.

In 1969–86, neither total university research expenditures nor the total number of degrees awarded are statistically significant in explaining variation in the MSAs' change in average earnings. The significant factors are as follows: location in the Midwest and West (negatively related), MSA size (positive), average earnings per job at the start of the period (negative), proportion of self-employed earnings (negative), change in per cent earnings from manufacturing over the period (positive), and presence of a large airport hub (positive). The results suggest that general regional macroeconomic conditions, agglomeration economies and aspects of industry structure rather than the presence or activities of universities were the most important factors determining gain in regional economic well being during this period. Where knowledge-producing activity mattered, it was *outside* universities.

The results are sharply different for 1986–98. Total university R&D activity is significant and positive, while the total number of degrees awarded is significant and negative. University patent activity, controlling for R&D expenditures, is non-significant. Location still matters, but now the advantage is only for MSAs in the Northeast. MSA size is still significant and positive, and average earnings per job at the start of the time segment are still significant and negative. The effect of industrial structure attributes has changed. The relative size of the business services sector – a measure of the development of the business infrastructure – is now significant and positive, but neither the relative size of the manufacturing sector nor of the self-employed sector is a significant factor. Private-sector patent activity is significant in this later period. The presence of a large airport hub also remains significant. The interaction between R&D and employment, however, is not significant in either period. Note that the overall explanatory power of the model in the second time segment is lower than in the earlier time segment, as indicated by the respective R^2.

CONCLUSIONS AND SUGGESTIONS FOR FURTHER RESEARCH

The first hypothesis, that research universities contribute significantly to regional economic development, controlling for other factors, was not supported by

Table 3. Multivariate regression model results, 1969–86

Dependent variable: change in average earnings, 1969–86
Total degrees of freedom = 294
Root mean square error = 5.35
$R^2 = 0.569$
Adjusted $R^2 = 0.542$
$F = 21.47$
$\Pr > F < 0.0001$

Variable	Parameter estimate	Standard error	t	$\Pr > t$	Tolerance
Intercept	34.70	3.81	9.10	<0.0001	
UNRD72	−0.01	0.05	−0.28	0.78	0.18
Degree72	−0.13	0.19	−0.66	0.51	0.14
West	−3.25	1.31	−2.48	0.01	0.37
Midwest	−4.59	1.11	−4.11	< 0.0001	0.41
South	1.73	1.11	1.56	0.12	0.33
Employ68	0.00	0.00	3.95	0.00	0.17
AVGEARN69	−0.26	0.04	−7.22	< 0.0001	0.40
MFG_EARN69	−0.05	0.03	−1.41	0.16	0.41
BUS_EARN69	−0.36	0.22	−1.62	0.11	0.76
Prop_Earn69	−0.80	0.11	−7.02	<0.0001	0.71
Co_Pats75	0.00	0.01	0.16	0.87	0.30
Univ_Pats75	0.13	0.27	0.49	0.62	0.74
CHMFG69_86	0.41	0.08	5.23	<0.0001	0.50
HunSm69	0.28	0.78	0.35	0.72	0.80
HubMed69	−1.14	1.19	−0.96	0.34	0.61
HubLrg69	3.40	1.45	2.34	0.02	0.43
RD72_x_Emp68	0.00	0.00	−1.68	0.10	0.23

Table 4. Multivariate regression model results, 1986–98

Dependent variable: change in average earnings 1986–98
Total degrees of freedom = 297
Root mean square error = 5.48
$R^2 = 0.458$
Adjusted $R^2 = 0.425$
$F = 13.92$
$\Pr > F < 0.0001$

Variable	Parameter estimate	Standard error	t	$\Pr > t$	Tolerance
Intercept	16.14	4.13	3.91	0.00	
UNRD86	0.04	0.01	3.44	0.00	0.21
Degree86	−0.41	0.15	−2.81	0.00	0.15
West	−3.88	1.20	−3.23	0.00	0.46
Midwest	−3.68	1.09	−3.38	0.00	0.45
South	−3.27	1.07	−3.05	0.00	0.37
Employ86	0.00	0.00	2.09	0.04	0.16
AVGEARN86	−0.23	0.04	−6.39	<0.0001	0.50
MFG_EARN86	0.03	0.04	0.85	0.40	0.54
BUS_EARN86	0.86	0.21	4.04	<0.0001	0.60
Prop_Earn86	−0.12	0.14	−0.87	0.39	0.66
Co_Pats86	0.01	0.00	2.97	0.00	0.20
Univ_Pats86	0.00	0.14	−0.00	1.00	0.47
CHMFG86_98	0.05	0.05	1.00	0.32	0.77
HubSm86	1.71	0.89	1.91	0.06	0.84
HubMed86	1.66	1.23	1.35	0.18	0.70
HubLrg86	6.84	1.41	4.85	<0.0001	0.44
RD86_x_Emp86	0.00	0.00	−1.86	0.06	0.22

the data throughout the full study. However, research universities did contribute strongly in the second period. In 1969–86, the presence or absence of a top 50 research university did not affect the gain in average earnings per worker, while it is a significant factor in the 1986–98 period. The explanation is that economic development activities were generally absent from universities pre-1986 and quite prevalent in the latter period. In addition, the latter period encompasses a more knowledge-based economy.

Therefore, the second hypothesis, that it is the universities' economic development activities that matter most, is confirmed by the data, but with one caveat. That after taking into account total university R&D activity (which is significant in 1986–98), university patenting does not appear to make a difference. This suggests that the mechanisms by which university R&D activity stimulates economic development are much broader and diverse than just patenting and licensing activity.

The third hypothesis is that the human capital creation and milieu functions of the university are important contributors to regional economic development. Since these factors are present throughout the two periods, and the university was not a significant factor in the earlier period, it appears these functions are not as important as had been originally thought. That MSAs with research universities outperform MSAs without top research universities in the age of the knowledge-based economy and the 'entrepreneurial university', controlling for other factors, is expected. That the presence of universities did not matter either way in 1969–86 supports the view that the teaching and milieu functions are not as important as the research and economic development functions of universities, since the former functions did not change appreciably over the full period, while research activity and economic development increased dramatically from the early to the later period. Indeed, the negative relationship between the number of degrees awarded by all institutions of higher education in the MSA and the dependent variable for 1986–98 adds further support to this view. The negative relationship can be interpreted as a tendency toward an over supply, or saturation, of highly educated workers in the average regional labour market. Since there is no indicator of the milieu function independent of teaching activity, unfortunately one cannot separate these factors when interpreting the results.

The fourth and fifth hypotheses, the questions of whether the agglomeration economies are more important than research universities, and whether the research universities can serve as a substitute for agglomeration economies, have been debated in the literature for some time. The present results provide mixed evidence. On the one hand, in 1986–98, the only MSA size category for which the presence of research universities made a significant difference was small MSAs, suggesting a substitute effect. In addition, the interaction variable turned out not to be significant in either period (at $p = 0.05$). On the other hand, the coefficient estimate for MSA size in the regression model for both periods indicates that agglomeration matters, independent of the presence and the size of the university sector. And the sign on the interaction variable, significant at $p = 0.10$, was negative for both periods. It might be that both these seemingly contradictory results are true; research universities in small areas can provide a number of external benefits that urban agglomerations generally provide.

Finally, although the presence of research universities and their scale of research activity are statistically significant factors in explaining gains in average earnings per job among MSAs in the later period, the strength of the causal relationship is modest. Controlling for other factors, it would have taken an increase of US$10 million in research expenditures among universities in an 'average' MSA to increase the index of average earnings per job by 0.36. To give these numbers some perspective, the average MSA had US$30.7 million in R&D expenditures in 1986. If the universities in this hypothetical MSA had increased their R&D expenditures by US$10 million more (about a 33% increase), the MSA would have increased its index from 100.00 to only 100.36.

Overall, the results provide additional support to the view that universities' research and technology development activities generate significant knowledge spillovers that are captured within the regional environment, and result in enhanced regional economic development. Yet, the magnitude of the contribution that universities' research and technology development activities play is small compared with other factors.

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NOTES

1. Other knowledge-producing organizations include federal laboratories and the full array of corporate and non-profit research and development (R&D) laboratories. In terms of total US R&D expenditures, however, universities comprised 14.0% in 1998 (National Science Board, 2000).
2. A third measure of innovation used is R&D intensity. It is measured by the number or percentage of R&D employees. Keilbach (2000), when testing his model on 327 *kreise* (districts) in West Germany, used R&D employment as a proxy for R&D intensity.
3. Five MSAs were excluded from the present study population as outliers due to their size or physical isolation: Anchorage (Alaska), Chicago (Illinois), Honolulu (Hawaii), Los Angeles (California) and New York (New York).
4. National Science Foundation, CASPAR database, 2001.
5. The size categories chosen for 1969 are: (1) under 75 000 employment, (2) 75 000–749 999 and (3) $\geqslant$750 000. The size categories for 1986 are: (1) $<$250 000, (2) 250 000–999 999 and (3) $\geqslant$1 000 000. This yielded 150, 147 and 15 MSAs, respectively, in the three size categories in 1969, and 227, 64 and 21 MSAs, respectively, in 1986.
6. In a number of cases, the n of one of the groups was small, a condition that often leads to unequal group variances. In these cases, the approximate t-statistic was used (SAS Institute, 1991).

REFERENCES

Acs Z. J., Audretsch D. B. and Feldman M. P. (1994) R&D spillovers and recipient firm size, *Review of Economics and Statistics* **76**, 336–340.

Arrow K. J. (1962) The economic implications of learning by doing, *Review of Economic Studies* **80**, 155–173.

Atkinson R. (1989) Some states take the lead: explaining the formation of effective and ineffective state science and technology policies. Unpublished PhD dissertation, University of North Carolina at Chapel Hill.

Brett A. M., Gibson D. V. and Smilor R. W. (Eds) (1991) *University Spin-off Companies: Economic Development, Faculty Entrepreneurs and Technology Transfer.* Rowman & Littlefield, Savage.

Cook T. D. and Campbell D. T. (1979) *Quasi-experimentation: Design & Analysis Issues for Field Settings.* Houghton Mifflin, Boston.

Feller I. (1990) Universities as engines of R&D-based economic growth: they think they can, *Research Policy* **19**, 335–348.

Felsenstein D. (1994a) The university in local economic development: benefit or burden? Unpublished paper, Department of Geography and Institute of Urban and Regional Studies, Hebrew University, Jerusalem.

Felsenstein D. (1994b) Dealing with 'induced migration' in university impact studies. Unpublished paper, Department of Geography and Institute of Urban and Regional Studies, Hebrew University, Jerusalem.

Fujita M., Krugman P. and Venables A. J. (1999) *The Spatial Economy: Cities, Regions, and International Trade.* MIT Press, Cambridge, MA.

Goldstein H. A. (2002) Universities as regional growth engines: the case of the Research Triangle of North Carolina, in Varga A. (Ed.) *Innovation, Entrepreneurship, and Regional Economic Development: International Experiences and Hungarian Challenges.* University of Pecs Press, Pecs.

Goldstein H. A. and Luger M. I. (1992) *Impact Carolina: The University of North Carolina at Chapel Hill and the State's Economy.* Final report prepared for the University of North Carolina at Chapel Hill Bicentennial Observance.

Goldstein H. A. and Luger M. I. (1993) *The Economic Impact of North Carolina State University's College of Engineering on the State.* Final report prepared for the College of Engineering, North Carolina State University.

Goldstein H. A., Maier G. and Luger M. I. (1995) The university as an instrument for economic and business development: U.S. and European comparisons, in Dill D. and Sporn B. (Eds) *Emerging Patterns of Social Demand and University Reform: Through a Glass Darkly.* Pergamon, Oxford.

Griliches Z. (1979) Issues in assessing the contribution of research and development to productivity growth. *Bell Journal of Economics* **10**, 92–116.

Hall B. H., Jaffe A. and Trajtenberg M. (2000) Market value and patent citations: a first look. Working paper of the National Bureau of Economic Research #7741.

Isserman A. (1987) An automated quasi-experimental control group method and its application to a tourism economic development project. Paper presented at the North American Regional Science Association Meetings, Baltimore, MD, USA, 6 November 1987.

Isserman A. and Merrifield J. (1987) Quasi-experimental control group methods for regional analysis: an application to an energy boomtown and growth pole theory, *Economic Geography* **63**, 3–19.

Jaffe A. (1989) Real effects of academic research, *American Economic Review* **79**, 957–970.

Jaffe A. and Trajtenberg M. (1996) Flows of knowledge from universities and federal laboratories: modeling the flow of patent citations over time and across institutional and geographic boundaries, *Proceedings of the National Academy of Sciences, USA* **93**, 12671–12677.

Jaffe A. B., Trajtenberg M. and Henderson R. (1993) Geographic localization of knowledge spillovers as evidenced by patent citations, *Quarterly Journal of Economics* **108**, 577–598.

Keilbach M. (2000) *Spatial Knowledge Spillovers and the Dynamics of Agglomeration of Regional Growth.* Physica, Heidelberg.

Ko S.-C. (1993) The incidence of high technology start-ups and spin-offs in a technology-oriented branch plants complex: the case of the Research Triangle region, North Carolina. Unpublished PhD dissertation, University of North Carolina at Chapel Hill.

Krugman P. (1991) *Geography and Trade.* MIT Press, Cambridge, MA.

Luger M. I. and Goldstein H. A. (1991) *Technology in the Garden: Research Parks and Regional Economic Development.* University of North Carolina Press, Chapel Hill.

National Science Board (2000) *Science & Engineering Indicators – 2000.* NSB-00-1. National Science Foundation, Arlington.

Reed W. R. and Rogers C. (2003) A study of quasi-experimental control group methods for estimating policy impacts, *Regional Science and Urban Economics* **33**, 3–25.

SAS Institute (1991) *SAS/STAT User's Guide, Release 6.03 Edition.* SAS Institute, Inc., Cary.

Saxenian A. (1994) *Regional Advantage: Culture and Competition in Silicon Valley and Route 128.* Harvard University Press, Cambridge, MA.

Shadish W. R., Cook T. D. and Campbell D. T. (2002) *Experimental and Quasi-experimental Designs for Generalized Causal Inference.* Houghton Mifflin, Boston.

Smilor R. W., Gibson D. V. and Dietrich G. B. (1990) University spin-out companies: technology start-ups from UT-Austin, *Journal of Business Venturing* **5**, 63–76.

Solow R. M. (1957) Technical change and the aggregate production function, *Review of Economics and Statistics* **39**, 312–320.

Steffensen M., Rogers E. M. and Speakman K. (2000) Spin-offs from research centers at a research university, *Journal of Business Venturing* **15**, 93–111.

THANKI R. (1999) How do we know the value of higher education to regional development?, *Regional Studies* **33**, 84–89.

THURSBY J. G. and THURSBY M. C. (2000) Who is selling the ivory tower? Sources of growth in university licensing. Working paper of National Bureau of Economic Research #7718.

VARGA A. (2000) Local academic knowledge transfers and the concentration of economic activity, *Journal of Regional Science* **40**, 289–309.